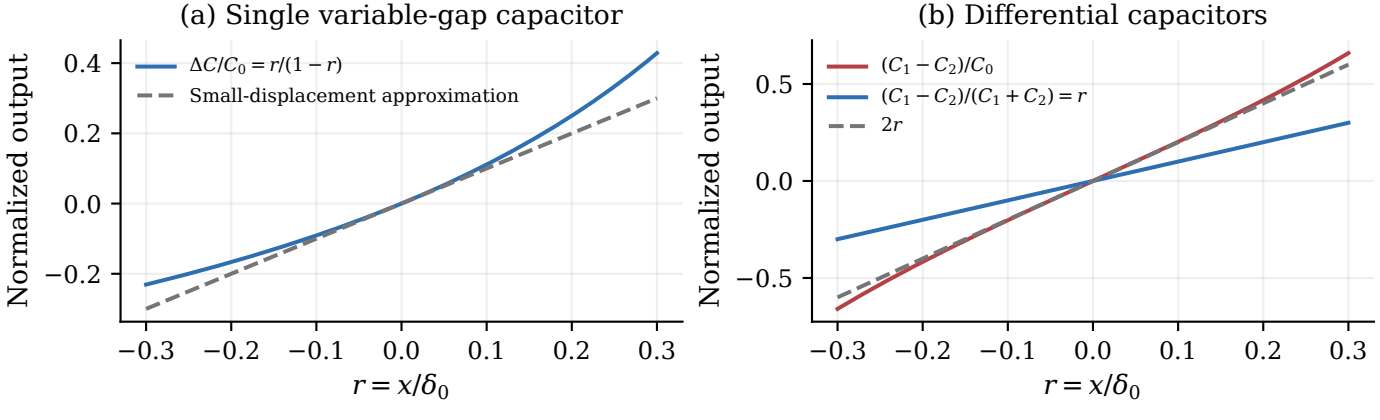
(a) Single variable-gap capacitor
Normalized output
$\Delta C/C_0 = r/(1-r)$
Small-displacement approximation
0.4
0.2
0.0
−0.2
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
$r = x/\delta_0$
(b) Differential capacitors
Normalized output
$(C_1 - C_2)/C_0$
$(C_1 - C_2)/(C_1 + C_2) = r$
$2r$
0.5
0.0
−0.5
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
$r = x/\delta_0$